

R Code

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1
2
3 # ==== Objective 1: To understand and perform basic operations for handling vectors,
  matrices, and data frames in R.
4
5         ## Vector
6
7 # Which are used to hold multiple data values of the same type is called vector
8 # One major key point is that in R the indexing of the vector will
9 # start from '1' and not from '0'. We can create numeric vectors and
10 # character vectors as well.
11
12
13
14         ## Types of Vectors
15
16 # Vectors are of different types which are used in R.
17 # Following are some of the types of vectors:
18
19 # (A) Numeric vectors
20
21 # Numeric vectors are those which contain numeric values such as integer etc.
22
23
24 # Example
25
26 x <- c(14, 15, 16, 17) # creation of vectors using c() function.
27 x
28
29 typeof(x) # display type of vector
30
31 class(x) # display class of vector
32
33
34
35 #(B)Character vectors
36
37 # Character vectors contain alphanumeric values and special characters.
38
39 x <- c('hi', 20, 'hello', 7)
40 typeof(x) # Displaying type of vector
41 class(x) # display class of vector
42
43
44 #(C)Logical vectors
45
46 # Logical vectors contain boolean values such as TRUE, FALSE and NA for Null values.
47
48 x <- c(TRUE, FALSE, TRUE, NA)# Creating logical vector using c() function
49
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50 typeof(x) # Displaying type of vector
51 class(x) # display class of vector
52
53
54
55 ### Creating a vector
56
57
58 x <- c(61, 4, 21, 67, 89, 2) # 1.Using c() function
59 x
60 y<- seq(from=1, to=15, by = 0.5) # 2. Using seq() function
61 y
62 z <- 2:27 # 3. Using ":"
63 z
64
65 ## Accessing vector elements
66
67 # Accessing elements in a vector is the process of performing operation
68 # on an individual element of a vector. There are many ways through which
69 # we can access the elements of the vector.
70 # The most common is using the '[]', symbol.
71
72
73
74 X <- c(2, 15, 18, 10, 12)
75 X[4] # 1. Accessing elements with an index number.
76
77
78 Y <- c(4, 8, 12, 11, 17)
79 Y[c(4, 1)] # 2. Using c() function
80
81
82
83 Z <- c(15, 21, 11, 41, 14, 31)
84 Z[Z>11] # 3. Using logical expressions
85
86
87
88
89 ## Modifying a vector
90
91 X <- c(2, 7, 9, 7, 8, 2)
92
93 X[3] <- 1
94 X[2] <-9 # 1. Modify a specific element
95 X
96
97 y<- c(2, 7, 9, 7, 8, 2)
98 y[y>5] <- 10 # 2.Modify using different logics.
99 y
100
101 z <- c(2, 7, 9, 7, 8, 2)
102 z[c(3, 5, 1)]<-22 # 3.Modify by specifying the position or elements.
103 z
```

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104
105
106
107 ## Sorting elements of a Vector
108
109
110 #sort() function is used with the help of which we can sort the
111 #values in ascending or descending order.
112
113 z <- c(8, 2, 7, 1, 11, 2)
114 I=sort(z)                # 1. Sort in ascending order
115 I
116 D=sort(z, decreasing = TRUE)
117 D
118
119 # 2. Sort in descending order
120
121
122
123 ## Combine more than two vector into a single vector
124
125 x=c(1,2,3)
126 x
127 y=c(4,5,6)
128 y
129 z=c(x,y) # Using c()function
130 z
131
132
133
134
135 ## To check the existence of an vector
136
137 cc <- c(12, 7, 3, 3, 8, 2)
138 is.vector(cc) # Command to check
139
140
141 ## Convert an object into a vector
142 v=1:9
143 v
144 m=matrix(v,nrow=3,ncol=3)
145 m
146
147 as.vector(m) # Command to convert in vector
148
149
150
151
152 ## Practice Question
153
154 # Question 1. Generate a vector from 10 to 40 by increment 0.5 and find the following:
155
156 #(a) Check the type of vector.
157 #(b) Arrange the vector in decreasing and increasing order.
```

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158 # (c) Extract the 4th and 8th term from the vector.
159 # (d) Change the 5th, 10th and 7th term with 100,101 & 120.
160 # (e) Discuss descriptive measures for vector (d).
161
162 # Question 2. Write the command for the following work in R
163
164 # (a) To quite R.
165 # (b) To delete whole console window.
166 # (c) To install the package.
167 # (d) To call the package from library.
168 # (e) To find out the descriptive measures of data.
169 # (f) To combine more than two vectors into a single vector.
170 # (g) To run the code in console window.
171 # (h) Path to arrange window.
172 # (i) Path to change font size.
173 # (j) To convert an object into a vector.
174 # (k) To check for the existence of a vector in R.
175 # (l) To find out the no. of observation in a given vector.
176 # (m) To get the help from console window.
177
178
179 ## =====
180
181
182 ## Matrix
183
184 # (A) Create a matrix with the help of given data.
185
186 data=c(2,5,6,8,9,10,12,13,21,22,23,12,44,55,11,10)
187 data
188
189 mat=matrix(data,nrow=4,ncol = 4) # command to create matrix
190 mat
191
192 mat1=matrix(data,nrow = 4,ncol=4,byrow =TRUE) # command to create a matrix with row
wise elements.
193 mat1
194
195
196 # (B) Extracting row, column, and sub-matrix from a given matrix.
197
198
199
200 col1=mat[,1] # Extraction of the 1st column
201 col1
202
203 row1=mat[1,] # Extraction of the 1st row
204 row1
205
206 submat=mat[3,2] # Extraction of the element of 3rd row and 2nd column
207 submat
208
209 submat1=mat[1:3,3:4] # Extraction of the sub-matrix
210 submat1
```

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211
212
213
214 # (C) Multiplication, transpose, and inverse of the matrix
215
216
217
218 A=mat # 1st matrix
219 B=mat1 # 2nd matrix
220
221
222 C=A%*%B # multiplication of the matrix
223 C
224
225 Inv_A=solve(A) # inverse of the matrix
226 Inv_A
227
228 Trps=t(A) # transpose of the matrix
229 Trps
230
231
232
233 # (D) Changing the row, and column names of the matrix
234
235 rownames(mat) <- c("R1", "R2", "R3", "R4") # command to change the row names of the
matrix
236 mat
237
238
239 colnames(mat)<-c("C1", "C2", "C3", "C4") # command to change the column names of the
matrix
240 mat
241
242
243
244
245
246
247 # Practice Question:
248
249 # Question 1. Create a vector of length 25 and find the following:
250
251 #(a) Convert data into a matrix.
252 #(b) Find the sub matrix of order 4*4, and 3*2.
253 #(c) Change the row names of matrix (a) by Row1, Row2, ... Row5.
254 #(d) Find the inverse, and transpose of the matrix (a).
255 #(e) Extract 5th column of the matrix.
256 #(f) Extract 2nd row of the matrix.
257 #(g) Create another matrix of order 5*5 and multiply with the given matrix (a).
258
259 # Question 2. Create an identity matrix of order 3*3.
260
261 # Question 3. Write the matrix for the given data and discuss
262 # descriptive measures for 3rd row of the matrix.
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263
264 # (data: 11,2,3,4,5,3,5,2,1,5,6,10,21,2,22,10,33,32,31,30,25,26,27,14,15)
265
266
267 ## =====
268
269 ## Data frame
270
271 # Variables in a data frame may be numeric (numbers) or categorical (characters or
272 # factors).
273 # In a data frame, we can combine variables of equal length.
274
275
276 #(A) Create data frame with the help of given vector
277
278 x=letters[1:25]
279 x
280 y=1:25
281 y
282 z=seq(11, 35,1)
283 z
284
285
286 data=data.frame(x,y,z) # command to make data frame
287 data
288
289
290
291
292
293 #(B) Extract variable from given data frame
294
295 d1=data.frame(x,y,z)
296 d1
297
298
299 vector1=d1$x      # "$" sign is used to extract variable
300 vector1
301
302 vector2=d1$X1     # "$" sign is used to extract variable
303 vector2
304
305 vector3=d1$z      # "$" sign is used to extract variable
306 vector3
307
308
309 ## Practice Question:
310
311 # Question 1. An example of data frame "painters" is available in the package "MASS".
312 # Find the following for given "painters" data:
313
314 #(a) Understand the description of data with the help of command.
315 #(b) Print the data.
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316 # (c) Find the structure of data.
317 # (d) Extract each variable.
318 # (e) Find the descriptive measures for each variables.
319 # (f) Check which variable is numeric and which is factor.
320 # (g) Extract the name of painters and name of variables.
321 # (h) Extract the top six row of data.
322
323
324
325
326
327 ###Code
328
329 # (a)
330 ?painters
331
332 # (b)
333 library(MASS)
334 data=painters
335 data
336
337 # (c)
338 str(data) # The result of str gives the dimension as well as the name and type of each
           variable.
339
340 # (d)
341 colour=data$Colour
342 colour
343 sch=data$School
344 sch
345
346 # (e)
347 summary(painters)
348 summary(sch)
349
350 # (f)
351 typeof(sch)
352 typeof(colour)
353 class(colour)
354 class(data)
355
356 # (g)
357 rownames(painters)
358 colnames(painters)
359
360 # (h)
361 head(data)
362
363
364
365 ##Extract data from a data frame
366
367 # The data from a data frame can be extracted by using the matrix-style [row, column]
368 # indexing.
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369 # Example: Suppose we want to extract information on the first painter "Da Udine"
370 # on the variable "Composition" from the data set painters.
371 painters["Da Udine", "Composition"]
372
373
374 ## Attaching a data frame
375
376 #1. With a command attach() over the data frame,
377 #   the variables can be referenced directly by name.
378
379 attach(painters)
380 summary(School)
381 summary(Drawing)
382
383 ## Detaching of data frame
384
385 detach(painters) #The command detach() recovers the default setting.
386 summary(School)
387
388 ## Subsets of a data frame
389
390
391 #1. We can obtained a subset() of complete dataset
392
393 subset(painters, Composition <= 6)
394
395 subset(painters, School=='F')
396
397 #2. Uninteresting columns can be eliminated.
398 #   The third and the fifth column ( Colour and School) are not shown.
399
400 subset(painters, School=="F", select=c(-3,-5))
401
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